Data Structures & Algorithms

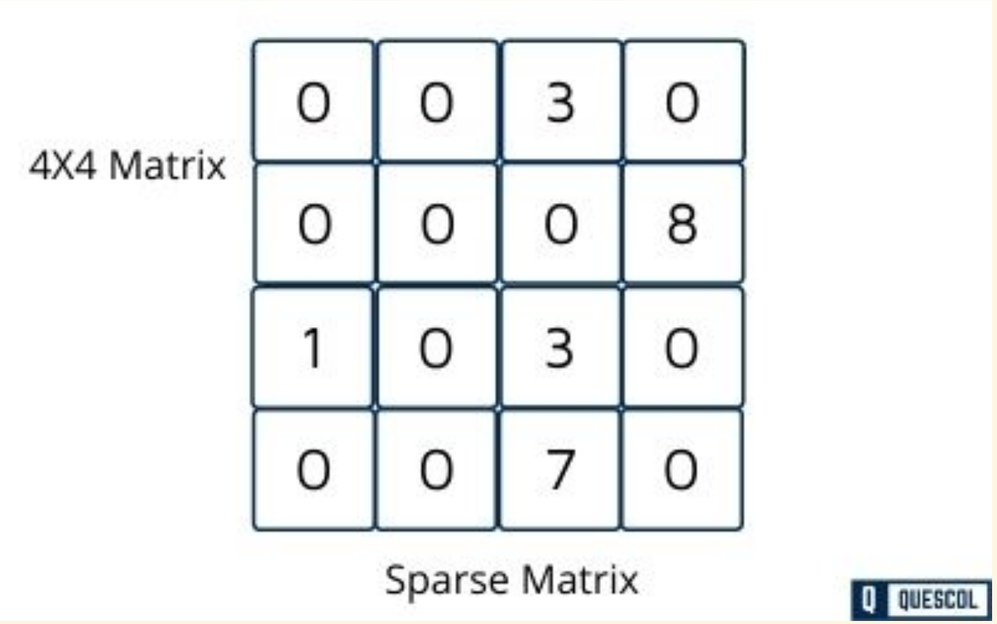
Practical Sessions

LAB SESSION #5

Explore sparse matrices and their applications

SPARSE MATRICES

- Sparse matrices are those matrices that have the majority of their elements equal to zero.
- Sparse matrix can be defined as the matrix that has a greater number of zero elements than the non-zero elements.



Why is a sparse matrix required if we can use the simple/dense matrix to store elements?

Because of the following two reasons.



SPARSE MATRICES

- **Storage** - We know that a sparse matrix contains lesser non-zero elements than zero, so less memory can be used to store elements. It evaluates only the non-zero elements.
- **Computing time:** In the case of searching in sparse matrix, we need to traverse only the non-zero elements rather than traversing all the sparse matrix elements. It saves computing time by logically designing a data structure traversing non-zero elements.

TYPES OF SPARSE MATRICES

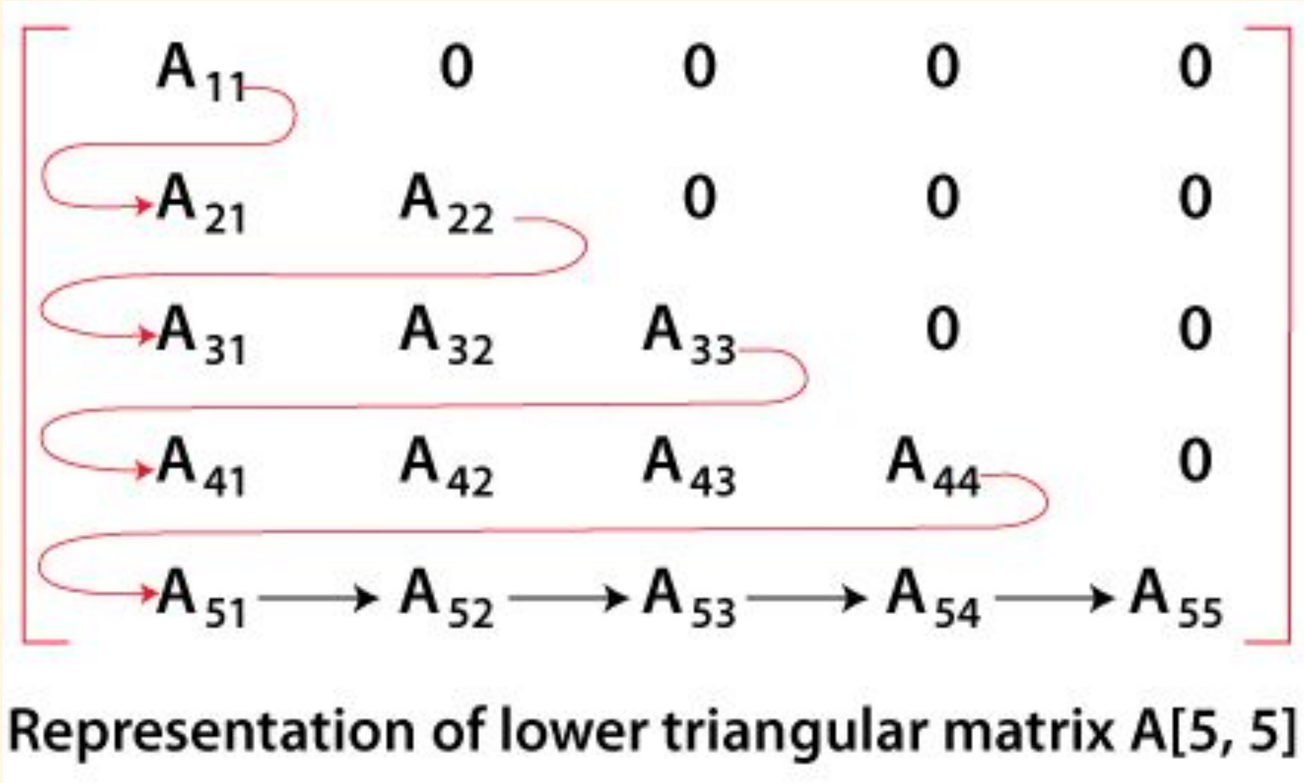
- **Lower triangular sparse matrices:**
  - A Lower regular sparse matrix is the one where all elements above the main diagonal are zero value.

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 9 & -2 & 0 & 0 & 0 \\ -2 & 1 & 3 & 0 & 0 \\ 3 & 1 & -1 & 6 & 0 \\ 0 & 6 & 7 & 2 & 7 \end{bmatrix}$$

Lower Triangular Matrix

TYPES OF SPARSE MATRICES

- **How to STORE Lower triangular regular sparse matrices?**
  - The 5 by 5 lower triangular regular sparse matrix as shown in the figure is stored in one-dimensional array B as:
  - B = { A11, A21, A22, A31, A32, A33, A41, A42, A43, A44, A51, A52, A53, A54, A55}



Representation of lower triangular matrix A[5, 5]

TYPES OF SPARSE MATRICES

- **How to STORE Lower triangular regular sparse matrices?**
  - To calculate the total number of non-zero elements, we need to know the number of non-zero elements in each row and then add them. If a sparse matrix has 1 non-zero entry in row 0, 2 non-zero entries in row 1, 3 nonzero entries in row 2, and n non-zero entries in row n-1. Total non-zero entries will determine the size of the uni-dimensional array
  - **1 + 2 + …… + i + …….. + (n-1) + n = n (n+1)/2**
  - Here **B [0] = A[0][0], B [1] = A[1][0], B [2] = A[1][1], and so on**

TYPES OF SPARSE MATRICES

- **If we need to store this…**
  - calculate the size first: (n=no. of rows)
  - **1 + 2 + ……. + i + …….. + (n-1) + n = n (n+1)/2 = 10**
  - **So the uni-dimensional matrix will have 10 entries**
  - Here **B [0] = A[0][0], B [1] = A[1][0], B [2] = A[1][1], and so on**

$$A = \begin{bmatrix} 4 & & & \\ 3 & -5 & & \\ 1 & 6 & 2 & \\ 8 & 0 & 5 & 9 \end{bmatrix}$$

Triangular Matrix

TYPES OF SPARSE MATRICES

- **Algorithm A5.1: STORETRIANGULAR(A)**
- **i=0**
- **For j = 0 to n-1**
  - **For k = 0 to j**
    - **U[i] = A[j][k]**
    - **i++**
  - **End for**
- **End for**

$$A = \begin{bmatrix} 4 & & & \\ 3 & -5 & & \\ 1 & 6 & 2 & \\ 8 & 0 & 5 & 9 \end{bmatrix}$$

Triangular Matrix

TYPES OF SPARSE MATRICES

- **If we need to Retrieve this…**
  - To retrieve A[J][K], we assume, U[L] = A[J][K]. L represents the number of elements in U up to and including A[J][K].
  - In the rows before row J, there are 1 + 2 + 3 + … + J = ½ * J * (J+1) non-zero numbers. In row J, there are K+1 elements upto and including A[J][K].
  - Since the array U has starting index 0, we need to subtract 1 from L.
  - L = ½ * J * (J+1) + K + 1 – 1 = ½ * J * (J+1) + K
  - So, U[½ * J * (J+1) + K] = A[J][K]

$$A = \begin{bmatrix} 4 & & & \\ 3 & -5 & & \\ 1 & 6 & 2 & \\ 8 & 0 & 5 & 9 \end{bmatrix}$$

Triangular Matrix

TYPES OF SPARSE MATRICES

- **Algorithm A5.2: RETRIEVETRIANGULAR(U)**
- 1. For j = 0 to n-1
- 2. For k = 0 to n-1
- 3. If k > j, A[j][k] = 0
- 4. Else A[j][k] = U[½*j*(j+1)+k]
- 5. End for
- 6. End for

$$A = \begin{bmatrix} 4 & & & \\ 3 & -5 & & \\ 1 & 6 & 2 & \\ 8 & 0 & 5 & 9 \end{bmatrix}$$

Triangular Matrix

TYPES OF SPARSE MATRICES

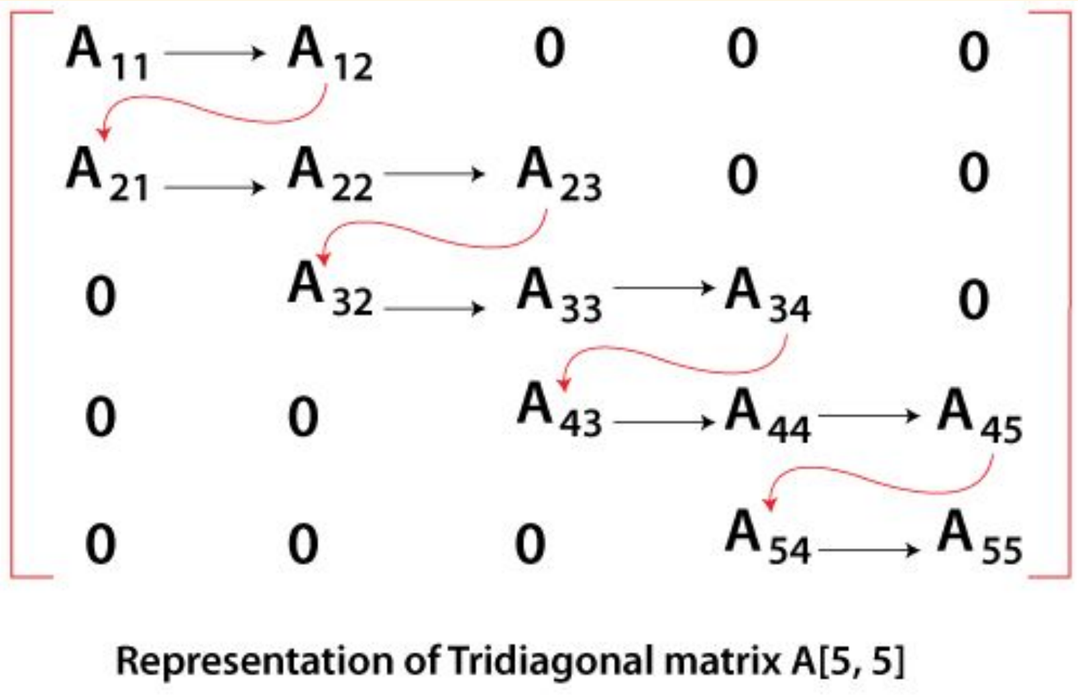
- **Tri-diagonal sparse matrix:**
  - The tridiagonal regular sparse matrix where all non-zero elements lie on one of the three diagonals, the main diagonal above and below.

$$\begin{bmatrix} 2 & 4 & 0 & 0 & 0 \\ 3 & -1 & 5 & 0 & 0 \\ 0 & 6 & 2 & 1 & 0 \\ 0 & 0 & 1 & -4 & 5 \\ 0 & 0 & 0 & 6 & 7 \end{bmatrix}$$

Tridiagonal Matrix

TYPES OF SPARSE MATRICES

- **How to STORE Tridiagonal sparse matrices?**
  - The 5 by 5 tri-diagonal regular sparse matrix, as shown in the above figure, is stored in one-dimensional array D as:
  - D = { A11, A12, A21, A22, A23, A32, A33, A34, A43, A44, A45, A54, A55}



Representation of Tridiagonal matrix A[5, 5]

TYPES OF SPARSE MATRICES

- **How to STORE Tridiagonal sparse matrices?**
  - In order to calculate the total number of non-zero elements, we need to know the number of **non-zero elements along the diagonal**, **above the diagonal** and **below the diagonal**.
  - The number of elements in a square matrix along the diagonal is **'n'** and above the diagonal is **(n-1)**.
  - So the total number of non-zero elements in n-square matrix is:

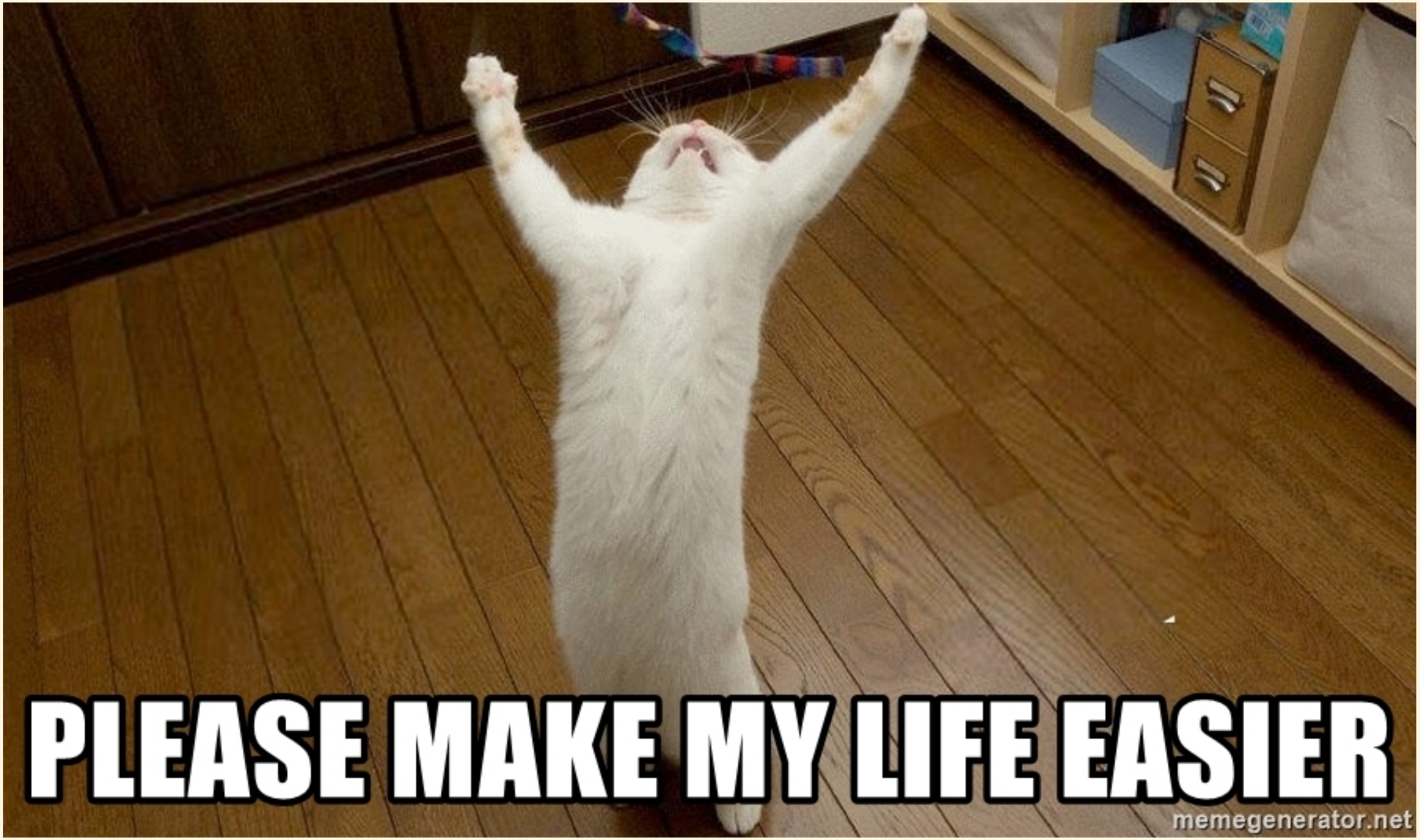
n	+	(n-1)	+	(n-1)
the diagonal		Above the diagonal		Below the diagonal

TYPES OF SPARSE MATRICES

- **How to STORE Tridiagonal sparse matrices?**

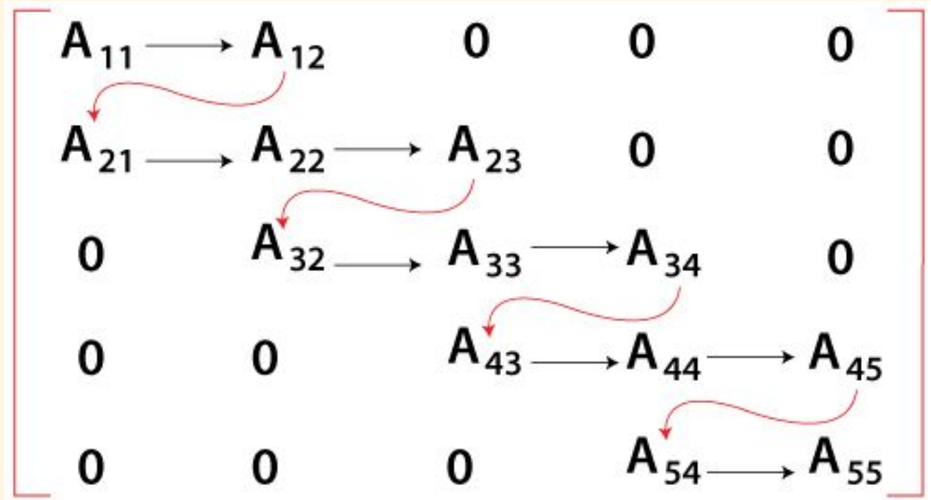
n	+	(n-1)	+	(n-1)
the diagonal		Above the diagonal		Below the diagonal

  -
  - Therefore, the total number of elements for n-squre matrix = 3n-2
  - In our case, n = 5 so the total number of elements = 3 * 5 - 2 = 13


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TYPES OF SPARSE MATRICES

- **How to STORE Tridiagonal sparse matrices?**
  - If we observe the indices of elements of:
  - **Main Diagonal:** row number is equal to column number (i = j).
  - **Lower Diagonal:** row number – column number = 1 (i – j = 1).
  - **Upper Diagonal:** row number – column number = -1 (i – j = -1).



Representation of Tridiagonal matrix A[5, 5]

TYPES OF SPARSE MATRICES

- **How to STORE Tridiagonal sparse matrices?**
  - They must be non-zero and all other elements must be zero.
  - We can combine these three conditions and write them as:

    |i – j| <= 1.
  - That is the absolute value of (i – j) is less than or equal to 1 then those elements must be non-zero. Now we can say that:

Main Diagonal : i = j.
Lower Diagonal: i – j = 1.
Upper Diagonal: i – j = -1.

M[i][j] ≠ 0 if |i – j| <= 1
M[i][j] = 0 if |i – j| > 1

TYPES OF SPARSE MATRICES

- **How to STORE Tridiagonal sparse matrices?**
  - **How many non-zero elements are there?**

    n + n-1 + n-1 (n = no. of rows)

    Or 3n – 2
  - non-zero elements are there/size of uni-dimensional array

Main Diagonal : i = j.
Lower Diagonal: i – j = 1.
Upper Diagonal: i – j = -1.

M[i][j] ≠ 0 if |i – j| <= 1
M[i][j] = 0 if |i – j| > 1


NOW IS THE TIME FOR ACTION.